Calculus I

Computation problems #10

Topics:

- Trigonometric substitution
- Integrals of rational functions
- Improper integrals

Problem 1. Evaluate the integral

$$\int \frac{dx}{\sqrt{4+x^2}}.$$

Problem 2. Evaluate the integral

$$\int \frac{dx}{\sqrt{25x^2-4}}, \quad x > \frac{2}{5}.$$

Problem 3. Expand the quotient

$$\frac{t^2+8}{t^2+5t+6}$$

by partial fractions.

Problem 4. Use partial fractions to evaluate the integral

$$\int \frac{x^2+4x+1}{(x-1)(x+1)(x+3)}\,dx.$$

Problem 5. Perform long division on the integrand, write the proper fraction as a sum of partial fractions, and then evaluate the integral

$$I = \int \frac{2x^3-2x^2+1}{x^2-x}\,dx.$$

Problem 6. The integral converges. Evaluate it without using tables.

$$I = \int_0^1 \frac{dx}{\sqrt{1-x^2}}.$$

Problem 7. Use integration, the Direct Comparison Test, or the Limit Comparison Test to test the integrals for convergence. If more than one method applies, use whatever method you prefer.

$$\int_0^{\ln 2} x^{-2} e^{-1/x}\, dx.$$

Problem 8. Evaluate the limit $\lim_{x\to\infty} \int_{-x}^{x} \sin t\, dt$.

Problem 9. Evaluate the limit $L = \lim_{n\to\infty} \sum_{k=1}^{n} \ln \sqrt[n]{1+\frac{k}{n}}$ by identifying them with definite integrals and evaluating the integrals.

Problem 10 Evaluate

$$\int_{-\infty}^{\infty} \frac{dx}{1+x^2}.$$

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Answer to selected problems

1. $\ln\left|\frac{\sqrt{4+x^2}}{2}+\frac{x}{2}\right|+C$
2. $\frac{1}{5}\ln\left|\frac{5x}{2}+\frac{\sqrt{25x^2-4}}{2}\right|+C$
4. $\frac{3}{4}\ln|x-1|+\frac{1}{2}\ln|x+1|-\frac{1}{4}\ln|x+3|+C$
6. $\frac{\pi}{2}$
7. $e^{-1/\ln 2}$
10. π